



Developing and extending usability heuristics evaluation for user interface design via AHP

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Abstract

Although Nielsen's heuristics are the most renowned usability heuristics, there is still vagueness and a lack of clarity which can be improved upon through modification or extension. Thus, Nielsen's ten heuristics and the four criteria of the Usefulness, Satisfaction, and Ease of USE Questionnaire were merged and modified to form the 14 main heuristics of the proposed usability heuristic. The analytic hierarchy process (AHP) analysis of the USE Questionnaire enabled the study to identify the top three questions per criteria. These were converted into sub-heuristics along with the tips provided via Nielsen's updated heuristics (2020), forming a total of 42 sub-heuristics. After the sub-heuristics were established, the research conducted AHP pairwise comparisons of the main heuristics to extract priority based rankings that will guide evaluators regarding which heuristics require more attention. The results validate the purpose of this usability heuristic evaluation, since the experts identified more problems than they did with Nielsen's heuristics proving a higher level of effectiveness.

Keywords Human computer interaction · Usability · Heuristics evaluation · User interface · AHP

1 Introduction

Usability is becoming increasingly important and a focus of experts' continuous research. One of the most popular evaluation methods is heuristic evaluation, which Nielsen and Molich define as "a usability engineering method for finding the usability problems in a user interface design so that they can be attended to as part of an iterative design process" (Nielsen and Molich 1990; Nielsen 1994). There are various heuristic evaluation methods including Nielsen's ten heuristics, The Eight Golden Rules of Interface Design proposed by Ben Shneiderman (Shneiderman 1998) and The Essentials of Interaction Design by Alan Cooper (Cooper et al 2014). Nevertheless, the most widely used evaluation method in the field of usability of user interface design is Nielsen's ten heuristics established in 1990 (Chen and Macredie 2005; Paz et al 2014; Dowding and Merrill 2018). Nielsen's heuristics were designed for user interface

design to discover usability problems. There are many advantages that encourage researchers to use this evaluation method; low costs, no extensive planning needed, encourages people to use it and most importantly it can be used in early stages of the development process as well as the evaluation phase (Nielsen and Molich 1990).

Nielsen's ten heuristics are criticised as some studies suggest that they focus on the system literally, rather than how the user experiences the difficulties of the system (De Jong and Lentz 2006; Kamper 2002). Many researchers have therefore developed new heuristics based on Nielsen's heuristics (Moraes and Silveira 2006; Clarkson and Arkin 2007; Mtebe and Kissak 2015) to try solving issues related to Nielsen's heuristics. However, these heuristics also suffer from several issues such as being domain specific (Mtebe and Kissak 2015; Kumar et al. 2019) therefore not allowing it to be generally used in usability evaluation, or they suffer from a lack of validation (Hermawati and Lawson 2016), or no clarity on how the heuristics were added (Muhanna et al 2020). The flaws identified in Nielsen's heuristics and the respective modified and developed versions, motivated the researcher to extend these set of heuristics via addressing the abovementioned weaknesses.

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Nielsen's worldwide known heuristics have been unchanged since 1994, however, in 2020, Nielsen updated the original article by adding new examples, tips and links. This study will focus on the tips that Nielsen included as they provide more explanation and clarity to the ten heuristics. As a result, the researcher decided to assign these tips as sub-heuristics to the main heuristics to make it clearer to evaluators, thus providing a clear path to the type of issues they would need to identify. According to our knowledge, these proposed heuristics are the first to extend Nielsen's heuristics using the tips provided by Nielsen. Furthermore, Nielsen's heuristics will be extended by merging them with the USE Questionnaire criteria and converting them into usability heuristics with the help of AHP. To our knowledge, this is the first study to rank heuristics based on priority via AHP with the aim to guide evaluators as to which heuristics require more attention based on importance.

Lund (2001) designed the USE (Usefulness, Satisfaction, and Ease of use) Questionnaire by collecting ideas from literature review and experimentation. After users evaluated systems via a seven-point Likert rating scale, Lund identified a regular pattern which built the dimensions of his questionnaire; Usefulness, Satisfaction, and Ease of Use. However, Lund decided to split the latter dimension into Ease of Use and Ease of Learning. Each of the four identified dimensions has a different number of questions totalling 19 questions. The addition of the USE Questionnaire aims to compensate for the broadness of Nielsen's heuristics and provide a more specific and effective usability evaluation model to make it more usable and beneficial to evaluators.

Analytic Hierarchy Process, known as AHP was developed by Saaty (Wind and Saaty 1980) to help researchers, from many expert backgrounds identify priorities in their multi criteria and consistency ratios that help with the decision-making of science experts, businesses, or any other fields (Chang 1996; Brunelli 2014). This study implements this analytical technique to assign priority rankings to help evaluators focus on the most important aspects of heuristic evaluation.

Ultimately, this study decided to choose Nielsen's heuristics as the subject of research due to its reputation and the potential it seen in using Nielsen's new tips for each category as sub-guidelines to establish as part of Nielsen's heuristics. The main contribution of this research is that it creates clear and efficient usability heuristics for user interface design via the inclusion of sub-heuristics from Nielsen's tips and USE Questionnaire criteria, along with the assignation of priority rankings via AHP analysis. This enables evaluators to prioritise the areas that require more attention and are more integral to the success of user interface. Furthermore, this study fills the gap identified in

many other studies regarding the lack of validation in previously proposed usability heuristics. This cemented the proposed heuristics and ensured that the contribution is clear through the practice of addressing the flaws of previous evaluation methods and outlining the benefits of the new usability heuristic.

This paper is split into the following sections: literature review which discusses topics related to previous research, identifying their key findings, and their advantages and disadvantages. Methodology discusses the steps taken in every part of this research in depth and describes the methods that the researcher used to gather data. The results and discussion section analyses the data gathered and establishes our proposed heuristics with validation. Finally, the conclusion and future works summarises the research and its contribution to the field of usability heuristic evaluation, as well as the limitations of this research.

2 Literature review

The term usability is defined by ISO 9241 standard as "The extent to which a product can be used by specified users to achieve specified goals with effectiveness, efficiency, and satisfaction in a specified context of use." Simply speaking, how quickly and easily can users use the system, how to minimise mistakes during the use of the system, and the ability to remember how to use the system again (Nielsen 1999; Agarwal 2004). An increase in the number and types of problems can harm a system (Jeddi et al 2020; Benaida and Namoun 2018; Viitanen et al. 2011) and ultimately result in poor usability, which directly impacts the efficiency and the satisfaction of users (Metzger et al 2010; Benaida 2014; Farrahi et al. 2019).

Various usability guidelines and heuristics have been established during the last two decades, thus many heuristic evaluation methods are being used by researchers and field experts. One of the most popular methods are Nielsen heuristics and Schneiderman's Eight Golden Rules (Schneiderman 1998); these eight golden rules include: (1) Strive for consistency, (2) Enable frequent users to use shortcuts, (3) Offer informative feedback, (4) Design dialog to yield closure, (5) Offer simple error handling, (6) Permit easy reversal of actions, (7) Support internal locus of control, (8) Reduce short-term memory load.

Nielsen's ten heuristics are popular amongst researchers who have tried to develop these heuristics by modifying and extending them. Nielsen Heuristic evaluation does not concentrate on specific domains, instead it is a general guideline that can be used for different fields such as web design. Nielsen's ten heuristics include (1) Visibility of system status (feedback), (2) Match between system and the real world (metaphor), (3) User control and freedom

(navigation), (4) Consistency and standards (consistency), (5) Error prevention (prevention), (6) Recognition rather than recall (memory), (7) Flexibility and efficiency of use (efficiency), (8) Aesthetic and minimalist design (design), (9) Help users recognise, diagnose, and recover from error (recovery), (10) Help and documentation (help). Research demonstrates that despite the passage of time, Nielsen's heuristics are still applicable and voluntarily utilised by experts. The constructive comments state that although it is widely used, there is still space for development, and it has the potential to be much more useful.

Testing methods and inspection methods are types of usability evaluation methods. The latter finds problems via the inspection technique and the expertise of the inspectors, whereas testing methods finds problems through observation of the actual users (Otaiza et al 2010). This study conducts the usability inspection method (heuristic evaluation) based on the experts that evaluate the new heuristics and systems.

There are many studies that try to extend or modify several existing heuristics using different methods. Six stages are suggested to establish new usability heuristics (Rusu et al 2011). For specific applications these include an exploratory stage, a descriptive stage, a correlational stage, an explicative stage, a validation (experimental) stage, and a refinement stage. However, it's vague and unclear about when and how the proposed phases can be implemented, thus it needs explaining.

Paddison and Englefield (2004) mention two main components when trying to establish heuristics; research based method and evaluation based method. The former involves establishing heuristics from literature review analysis, whereas the latter involves the analysis of previous data research. However, there is no clarity in the method that must be undertaken for the evaluation based method. Readers may understand analysis of previous data research to include conducting primary research or carrying out a meta-analysis of other peoples' results. (Sim et al 2009).

The authors (Hermawati and Lawson 2016) believe that usability heuristics should be created by extracting information and ultimately converting them into heuristics. However, their study only investigates heuristics for specific domains for which they are criticised for their lack of validation thus it limits the ability to improve existing heuristics and inability to encapsulate the true potential of it.

Several authors have published web usability guidelines that have not been validated through interface implementation and testing. Bailey's research (Bailey 2017) focuses on the human-computer interaction (HCI) of teenagers between the ages of 13 and 17. Bailey proposes a model that employs empirical studies to investigate these

guidelines. The results, at least in terms of the usability variables that were tested, are compatible with the recommended usability factors and 'cool' guidelines. The results of this research do not represent any other age category and cannot be said to be globally representative of all teenagers.

Alonso-Virgós et al, (2020) examines compliance with web usability guidelines for providing information in a timely and understandable manner. He collated the behaviour of 20 inexperienced web usability engineers in order to analyse the data on the application and the compliance with each of the study's guidelines. Authors of this research identified a need for the availability of standardised guidelines that assist user experience on web systems.

Do Amaral et al, (2018) address the issue of extensive information content available to various people. However, universal accessibility and usability criteria are difficult challenges to overcome. Do Amaral and his team propose assisting the integrated evaluation by creating an environment for analysing the results of accessibility and usability assessments, thus combining both human and technological creations to produce improvements on both fronts. Burghardt et al, (2013) outline an evaluation of eight desktop search engines' (DSEs) usability. They used a heuristic walkthrough method to identify usability issues as well as particular strengths and weaknesses of the search engines under consideration. The findings of this evaluation have been incorporated into a set of 30 design principles for user-friendly DSEs.

According to Alomari et al. (2020) an essential measure of success is the process of evaluating the utility of a cyberlearning environment in supporting the design process and preserving user pleasure. Authors used cognitive walkthrough combined with think-aloud protocol and a heuristic evaluation survey to improve cyberlearning practise by emphasising the importance of analysing the environment surrounding a certain activity and its users using UI and UE assessments. Muhanna et al. (2020) provide a new set of heuristics for assessing the usability of systems with Arabic interfaces. The emphasis was on a formal heuristic evaluation as the inspection tool for discovering flaws in these systems. Muhanna and his group compared the findings of Nielsen's heuristics with those using the proposed heuristics on three Arabic interfaces. Results show that evaluators were able to detect more serious flaws than those who used Nielsen's heuristics.

Delice and Güngör (2009) implemented two steps that aim to identify as many usability issues as possible. Firstly, heuristic evaluation using Nielsen's heuristics to identify the issues, and then AHP to rate the severity. By merging these two methods, evaluators are more likely to build user-friendly systems. Nevertheless, Delice and Güngör used a traditional approach and did not modify or extend

the heuristics to help identify more usability issues. Allen et al. (2006) note that the straightforward approach of presenting screenshots of a user interface for evaluation is the most cost-efficient and effective. Furthermore, the issues identified after heuristic evaluation should be categorised to ensure that they are straightforwardly addressed.

Prascevic and Prascevic (2017) propose a new procedure for identifying alternatives in the Fuzzy Analytic Hierarchy Process (FAHP) with trapezoidal fuzzy numbers by employing a new method for determining eigenvalues and eigenvectors of the criteria and alternatives based on expected values of the fuzzy numbers, as well as determining the weights of criteria. Linear programming helps calculate the local and global fuzzy weights of the options.

The literature review conducted in this study reveals that the vast amount of proposed guidelines or heuristics do not validate their propositions by merging and modifying existing heuristics. However, validation illustrates how the proposed heuristics address weaknesses of previous heuristics and the benefits of establishing these new heuristics. Therefore, it is crucial to validate any proposed heuristic to clarify the reason for its introduction. Furthermore, both Nielsen's heuristics and the USE Questionnaire are commonly used in the world of usability, however there is still room for improvement and adjustments. Therefore, this study aims to demonstrate the advantage of combining these evaluation methods with the aim of providing a more efficient and effective method for evaluators. Table 1 summarises the methodologies used in previous research to develop or extend heuristics, along with their relevant advantages and disadvantages.

3 Methodology

The research methodology for establishing new usability heuristics framework in this study consists of several steps (Fig. 1). Data collected from literature review was used to determine the use of Nielsen's ten heuristics in our proposed set of heuristics and to establish new sub-heuristics for them. Participating experts performed heuristic evaluation of two selected systems using Nielsen's heuristics alone which will help in validating our proposed heuristics later. AHP analysis of the sub-criteria of the USE Questionnaire was performed to identify priority-based rankings that help eliminate undesired data and establish the desired data as sub-heuristics for the four main criteria of the USE Questionnaire. By merging this with Nielsen's heuristics and their respective sub-heuristics, the first version of our usability heuristics was finally formed. AHP analysis was performed once again to determine priority-based rankings of the first version of our proposed heuristics. This established the final version of our usability heuristics, which

was then validated by allowing participating experts to use it to evaluate the two selected systems. The results of this evaluation were compared with the initial heuristic evaluation via Nielsen's heuristics.

3.1 Data collection

Data collection through experts, known as heuristic evaluation, is implemented to perform thorough analysis of the system to point out issues that a normal user would usually be unable to identify. The purpose of heuristic evaluation is to identify any serious usability problems that users may encounter and to recommend improvements to the design of the system before its implementation (Tan et al 2009; Choi and Bakken 2010). Therefore, five experts aged between 37 and 60 from different specialisation backgrounds were gathered: two experts in Human Computer Interaction, two in Software Engineering and one in Information Systems (Table 2). Throughout this research, the experts are not substituted to ensure a fair test and evaluation of the involved criteria. Experts have prior experience with Microsoft Teams and Blackboard. However, to ensure internal testing validity, the researcher allocated extra time slots for experts to get familiar with the latest updates of the systems in case they haven't already. These experts provide insightful feedback to the systems tested in this study by way of evaluating two systems via Nielsen's ten heuristics, completing an AHP pairwise comparison of the USE Questionnaire independently and then our proposed heuristics, and finally evaluating the same two systems via our proposed heuristics to help with validation.

3.2 Experiment

This experiment was conducted for one day in early 2021 in an offline setting. Five experts gracefully accepted our invite to partake in this study for the sake of research. All five experts were invited by the researcher at the same day and time to undergo this experiment simultaneously in the library of the Islamic University of Madinah, nevertheless, each participant could only see their screen. The researcher overlooked the whole experiment to ensure everything proceeded smoothly. A small brief was given to the participants explaining the overview of our research and the details of the experiment they will partake in. The same laptop running on the same internet connection was provided for each expert to avoid any external factors that may influence the results of our research and to maintain internal instrumentation validity. In total, there are four main parts to the experiment that the experts took part in and no time limit was given. However, the experiment would not proceed to the next step until all participants

Table 1 Methodologies and shortcomings of previous studies

Name and year	Methodology	Advantages	Disadvantages
Muhanna et al (2020)	Explores cultural aspects during evaluation	Their heuristics found more problems than Nielsen's heuristics specific to Arabic interfaces	No clarity on how heuristics were added
Figueroa et al (2019)	Applied the Prometheus methodology to develop VLE heuristics	VLE heuristics for virtual learning environments are better in discovering more problems than Nielsen's heuristics	Only studied the results of the application of Prometheus as an indirect way to measure its effectiveness
Hermawati and Lawson (2016)	Analyses heuristics in specific domains and their route to establishment	Comprehensive review of 70 research papers	Limited to heuristics for specific domains only Results show lack of validation quality to prove that specific domain heuristics are useful
Huang and Benyoucef (2014)	Uses Nielsen's heuristics to discover link between usability and credibility on e-government websites. When usability was strong, credibility was strong and vice versa	Focuses on the relationship between usability and credibility on government websites	The participants have different skills which affects the evaluation of the websites. They examine only three e-government websites
Bowie (2012)	Established seven heuristics and 127 guidelines specific to podcasts	One of the few studies to analyse podcasts or even audio	No validation
Rusu et al (2011)	Six stages for developing new heuristics	Accommodates for emerging applications	Lack of clarity in some stages Not applicable for existing applications or general usability
Delice and Güngör (2009)	Use heuristic evaluation to identify issues and AHP to rank severity of issues found	Applies severity ranking of problems	Uses dependent variables in AHP which assumes input variables to be independent Ranked the problems found rather than the heuristics
Paddison and Englefield (2004)	Research based method and evaluation based method for establishing heuristics	Encourages to educate evaluators before evaluating heuristics for more beneficial results	Analysis of previous data research method not specified

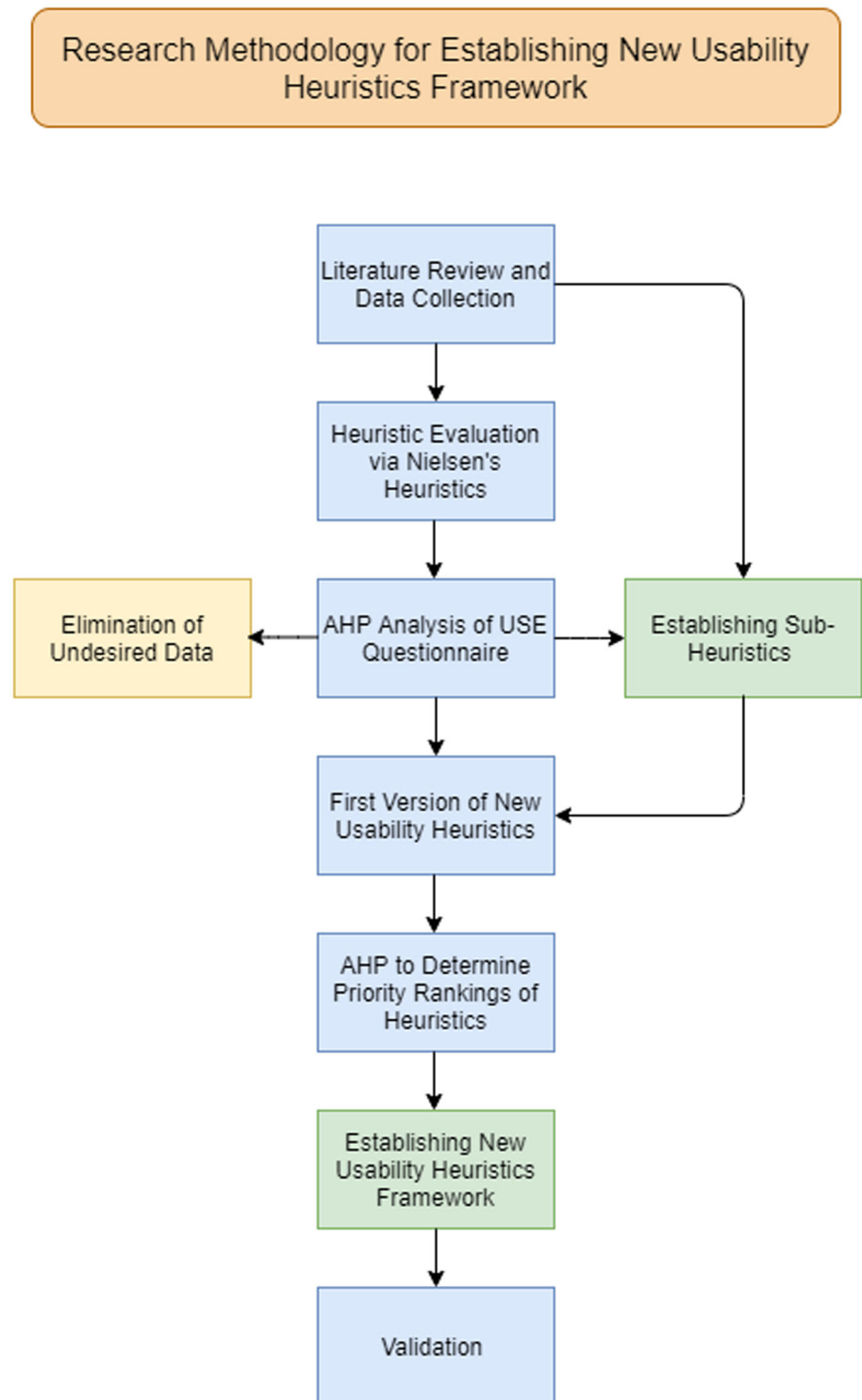
were finished. In the first part of the experiment, experts were provided with Nielsen's heuristics to evaluate Microsoft Teams and Blackboard, along with a sheet of paper which included the tasks that had to be completed during their evaluation to help identify errors in the selected systems. The same tasks were assigned across both selected systems. There are five main tasks that the experts had to complete; find and access the grade centre page, find the students timetable, find the course material, upload an empty file to the assessment page, finally, send an email to the tutor. In the second part of the experiment, experts had to complete AHP pairwise comparisons for the sub-categories of the four main categories of the USE Questionnaire (Usefulness, Ease of Use, Ease of Learning, Satisfaction) on the AHP link provided by the researcher. This would allow the researcher to exclude undesired data and keep three sub-categories based on importance that will ultimately feature in the new set of heuristics. After that, the experts had to undergo AHP pairwise comparisons for the first version of the proposed heuristics so that the

researcher can establish priority rankings which would finalise our proposal. Finally, the experts were provided once again with the same sheet of tasks as in the first part of the experiment, so that they could evaluate Microsoft Teams and Blackboard. However, this time they would evaluate the systems using our final version of the proposed heuristics. The results of their evaluation in the final part of the experiment were compared to the results of their evaluation in the first part of the experiment to confirm if our new heuristics helped identify more issues.

3.3 AHP

AHP is an analytical technique used to provide a hierarchical structure of selected data that can be completed by anyone, from normal users or customers to experts and business managers. Data input includes subjective and objective variables such as criteria for evaluators, environmental traits, and many other entities. It is mainly based on pairwise comparisons of the multi criteria investigated

Fig. 1 Research methodology for establishing new usability heuristics framework



since it compares each criterion individually with every other criterion via a nine-point scale. As seen in Table 3 (Saaty 1987), the scale ranges from 1 which represents “Equally Important Preferred” up until 9, which represents “Extremely Important Referred”. To find out the number of comparisons you will end up with, there is a simple

mathematical formula which can be used and is known as Gauss formula. Gauss discovered this mathematical method in the early nineteenth century. It helps to determine the sum of a series being investigated and is calculated using an algebraic formula where n represents the number of variables that the user inputs into the AHP

Table 2 Demographic information of experts

Experts	Age	Gender	Speciality
1	41	Male	Software Engineering
2	37	Male	Software Engineering
3	60	Male	Human Computer Interaction
4	52	Male	Software Engineering
5	49	Male	Data Mining

analysis (Saaty 2016, 2008; Bhushan and Rai 2007; Schmidt et al. 2015).

$$\text{Gaussformula : } \frac{n(n-1)}{2}$$

AHP analysis usually constitutes three levels; the main goal, criteria and alternatives. Nevertheless, it can be completed with the first two levels alone. The ultimate goal is to rank the variables in accordance to priority, based on the weighted criteria used to complete this analytical process. Consistency ratio (CR) determines how consistent the experts are in their judgement of the variables tested. The analysis is calculated via a formula which consists of dividing the consistency index (CI) gathered from the pairwise comparisons by the index for the corresponding random matrix (RI).

$$CR = \frac{CI}{RI}$$

$$CI = \frac{\lambda_{\max} - n}{n - 1}$$

$$CR = \frac{RI(\lambda_{\max} - n)}{n - 1}$$

Saaty (1987) determined that if the consistency ratio value is more than 10%, the data is unreliable due to a lack of consistency in the data gathered. On the other hand, if the value is less than 10%, they are more reliable and it displays a good level of consistency within the data. The closer the value is to 0, the more consistent the data is.

AHP group consensus tries to determine if the opinions of the experts are in line with each other or if they disagree with one another. The values are calculated as percentages, the higher the value, the larger the level of consensus between the experts. Any value above 85% indicates a very high level of consensus, whereas any value below 50% indicates a very low level of consensus.

3.4 AHP analysis in this research

This study uses AHP for two main reasons, firstly, to identify the three most important sub-criteria of the four categories in the USE Questionnaire, and secondly, to compile a priority ranking in descending order of importance of the proposed heuristics. These will measure the level of problems on the tested systems and will help us to

Table 3 The fundamental scales for pairwise comparison. (Saaty 1987)

Intensity of importance on an absolute scale	Definition	Explanation
1	Equal importance	Two activities contribute equally to the objective
3	Moderate importance of one over another	Experience and judgement strongly favour one activity over another
5	Essential or strong importance	Experience and judgement strongly favour one activity over another
7	Very strong importance	An activity is strongly favoured and its dominance demonstrated in practice
9	Extreme importance	The evidence favouring one activity over another is of the highest possible order of affirmation
2, 4, 6, 8	Intermediate values between the two adjacent judgements	When compromise is needed
Reciprocals	If activity i has one of the above numbers assigned to it when compared with activity j, then j has the reciprocal value when compared with i	
Rationals	Ratios arising from the scale	If consistency were to be forced by obtaining n numerical values to span the matrix

understand the severity of the issues within it. This process will enable evaluators to identify the best solutions for these issues and to minimise the number of problems that can affect the target systems.

AHP analysis of the USE Questionnaire is conducted through an online system (<https://bpmmsg.com/ahp/index.php>), which is a free web-based AHP solution. The primary aim of this OS is to complete the analytical steps for the user and generate the priority based ranking of the tested criteria to help users in their decision-making processes. The user must undergo several steps to set up an AHP project on the online system, which then allows the participants to fill out the pairwise comparison questionnaire. Session administration and project administration are the two main menus for the users' projects. The former allows the user to open an active project or start a new one, whereas the latter allows the management of a project. Once a user starts a new project, they are presented with a decision hierarchy input menu where the user must insert a decision matrix (or multiple). A name is assigned to the project along with a short description so that the participants who later fill out the questionnaire have a full understanding of the situation. The online system generates a link that participants can follow to fill out the pairwise comparisons of the variables included by the user. After each of the participants have completed the questionnaire, the user can analyse the results via the group results menu. The results are analysed thoroughly by the online system which displays multiple data analysis results. These include project result data, hierarchy with consolidated priorities, consolidated global priorities or weights of alternatives, sensitivity analysis, alternatives by participants, breakdown by nodes, and global priorities, otherwise known as hierarchy evaluation. Each category displays deep analysis providing users with all the necessary information they may need. Under the breakdown by nodes section, consistency ratio (CR) and the AHP group consensus are provided to the user (Goepel 2018).

3.5 Data categorisation and exclusion of undesired data

In 2020, Nielsen addressed the issue that his ten heuristics were very broad and often perceived as not specific enough, thus he included three tips to each category on his website. Consequently, a need to expand on Nielsen's heuristics is necessary due to the above-mentioned shortcomings (Nielsen 2020). The expansion of these heuristics in this study includes the addition of the tips mentioned as sub-heuristics in the new evaluation proposed to provide more clarity to the evaluators. This can help them identify more problems within the heuristic itself, thus proving to be more useful. Therefore, experts completed a heuristic

evaluation via Nielsen's heuristics, of the two most common software programmes in education, Microsoft Teams and Blackboard. The number of usability problems identified will be compared with the final heuristic evaluation conducted by the experts on the same software programs, however, this will be conducted using our proposed heuristics. This comparison aims to validate it and prove that it helps evaluators identify more problems within the systems because the heuristics are more specific and provide a clearer route for evaluators as a result of the addition of the tips classified as sub-heuristics. In the USE Questionnaire, Usefulness contains eight questions, Ease of Use contains 11 questions, Ease of Learning contains four questions, and finally, Satisfaction contains seven questions. To be consistent with the previous number of sub-heuristics gathered per heuristic, experts completed AHP analysis of the USE Questionnaire to allow the researcher to identify the three most influential sub-criteria. These sub-criteria will be included in the proposed heuristics, whereas the remaining sub-criteria do not contribute to the study any further, being classed as undesired data. Ultimately, the proposed heuristic at this stage adopts both Nielsen's ten heuristics and their sub-heuristics, along with the four categories of the USE Questionnaire and their respective sub-heuristics gathered via AHP analysis.

3.6 Expert evaluation of proposed usability heuristics

The study proposed this first version of the heuristic evaluation to the experts involved in this study as they completed an AHP pairwise comparison of all 14 heuristics to help rank the heuristics based on priority. The researcher decided not to include the sub-heuristics of the first ten heuristics in this pairwise comparison as they were tips gathered from Nielsen himself. In addition, the sub-heuristics from the USE Questionnaire have already undergone an AHP analysis and the priority rankings have been assigned. Therefore, the heuristics alone must undergo AHP analysis to complete the final step of establishing this evaluation model. According to Gauss formula, where n in this case is 14, the number of questions that the experts have to complete equates to 91. Consequently, the aim of this stage is to establish the correct arrangement of the heuristics in descending order of importance according to the data gathered from the AHP analysis, therefore evaluators will know which heuristic requires more attention.

3.7 Validation of proposed usability heuristics

The researcher proposes these usability heuristics as a general method that can be used in all fields. To further

cement and establish these proposed heuristics, the author decided to present it to the experts to evaluate Microsoft Teams and Blackboard as they are the most commonly used education systems. The results of this evaluation will be compared with the initial heuristic evaluation completed by the experts via Nielsen's heuristics on the same systems (Microsoft Teams and Blackboard). The comparison includes looking at the number of problems identified via Nielsen's heuristics compared to the proposed heuristics. The above-mentioned steps are essential in proving that they collectively enable evaluators to identify more problems within tested systems.

4 Results and discussion

4.1 Nielsen's Heuristic evaluation

Five experts conducted heuristic evaluation of Microsoft Teams and Blackboard using Nielsen's heuristics. Table 4 displays the number of issues identified across all of Nielsen's heuristics in both tested systems and their total. The values in bold indicate the total of the problems identified in Microsoft Teams (84), Blackboard (154), and both systems in accumulation (238). This will later be compared with the number of flaws identified via usability heuristics to verify the importance of including sub-heuristics. Experts identified 20 issues in Help users recognise, diagnose, and recover from errors, which is the highest number across any of Nielsen's heuristics in one system. Whereas, the lowest number of issues was in the Aesthetic and minimalist design department, four on Microsoft Teams and seven on Blackboard.

4.2 AHP of USE Questionnaire

AHP analysis of the USE Questionnaire allows the researcher to eliminate any undesired data to identify the three most important sub-heuristics per category (Usefulness, Ease of Use, Ease of Learning, Satisfaction) that will be used in the new usability heuristics. AHP analysis for each category is conducted independently since the goal, or otherwise known as level 0 in AHP terminology, is different each time. The data does not overlap with each other, so there are four separate tables explaining the AHP analysis results of the USE Questionnaire. Using Gauss formula, we can calculate the number of pairwise comparisons that the experts undergo. For usefulness, where n is nine, the number of pairwise comparisons is 36. Ease of use contains 55, ease of learning contains six, and finally, satisfaction contains 21 pairwise comparisons. The total number of pairwise comparisons for this section is 118 per expert. Furthermore, there is a total number of 30 questions in the USE Questionnaire, however only 12 of these questions will be utilised further in this research and the remaining 18 will be disregarded.

Eight questions were analysed in the category of usefulness. AHP analysis shows that the top three priority ranked criteria highlighted in bold, are "it is useful", "it helps me be more productive", and "it helps me be more effective", therefore, this was the data required for this usability heuristic. The remaining five questions, identified as undesired data (Table 5), were not used in this study any further. There is a large difference between the three desired criteria and all the other criteria in this category. The top three questions attained a priority score of 22.3%, 21.4% and 20.4%, respectively, whereas the fourth ranked attained only 9.4%. Therefore, there was no dispute between the experts in this section. The high score attained

Table 4 Number of problems identified by Nielsen's heuristics

Nielsen's Heuristics	Number of problems identified	
	Microsoft Teams	Blackboard
Visibility of system status	11	15
Match between system and the real world	10	18
User control and freedom	8	19
Consistency and standards	6	9
Error prevention	8	15
Recognition rather than recall	8	19
Flexibility and efficiency of use	10	15
Aesthetic and minimalist design	4	7
Help users recognise, diagnose, and recover from errors	9	20
Help and documentation	10	17
Total	84	154
	238	

Table 5 Priority ranked criteria of Usefulness category

Question	Priority	Rank	Desired or Undesired Data
It helps me be more effective	20.4%	3	Desired
It helps me be more productive	21.4%	2	Desired
It is useful	22.3%	1	Desired
It gives me more control over the activities in my life	9.4%	4	Undesired
It makes the things I want to accomplish easier to get done	5.9%	7	Undesired
It saves me time when I use it	8.7%	5	Undesired
It meets my needs	6.6%	6	Undesired
It does everything I would expect it to do	5.3%	8	Undesired

in this category for the AHP group consensus (90.0%) further supported this. The consistency ratio was almost perfect, as it gained 1.5%, thus the data is very reliable and consistent.

Ease of use contains 11 questions, which is the most of any category, consequently only 27% of the questions was considered desired data. Data analysis displays the priorities of the questions along with the consistency ratio and group consensus of the information gathered (Table 6). In order of rankings where the top three criteria are bold as seen in Table 6, “it is flexible” (20.5%), “it is easy to use” (18.1%), and “it is simple to use” (14.4%) are the questions included under the heuristic ease of use. This set of data attained more optimum scores than the previous section in both the consistency ratio (1.4%) and AHP group consensus (93.0%).

In the category of ease of learning, only one question is disregarded since there are only four questions in total. “I quickly became skilful with it” attained the lowest priority score of 15.3% and was eliminated. Whereas “it is easy to learn to use it”, “I learned to use it quickly”, and “I easily remember how to use it” attained priority scores of 39.3%, 25.2% and 20.3% respectively, as their rank is marked in bold in (Table 7). These questions were extracted and

merged with the proposed usability heuristic. The consistency ratio of this data was close to perfect (0.3%), however, although the AHP group consensus attained 78.6%, it is still considered to be high.

The final examined category was satisfaction, with a total of seven questions that were analysed via AHP analysis (Table 8). Through the analysis, this study identified “it is pleasant to use” (25.9%), “I am satisfied with it” (19.9%), and “it works the way I want it to work” (19.0%) as the top three priority ranked questions in this section, which is emphasised by their bold font in the rankings column. The experts were consistent in their judgements and thus the data is reliable (consistency ratio 1.1%). AHP group consensus (88.5%) shows that there was a high level of agreement amongst the judgements of the experts involved.

4.3 AHP of proposed usability heuristics

AHP analysis of these proposed heuristics allows the heuristics to be displayed in order of relative importance allowing evaluators to identify which aspects of the system require most attention. For this section, 14 variables undergo AHP analysis, thus there are 91 pairwise

Table 6 Priority ranked criteria of Ease of Use category

Question	Priority	Rank	Desired or Undesired Data
It is easy to use	18.1%	2	Desired
It is simple to use	14.4%	3	Desired
It is user friendly	11.7%	4	Undesired
It requires the fewest steps possible to accomplish what I want to do with it	5.5%	6	Undesired
It is flexible	20.5%	1	Desired
Using it is effortless	5.3%	7	Undesired
I can use it without written instructions	4.2%	9	Undesired
I don't notice any inconsistencies as I use it	3.5%	11	Undesired
Both occasional and regular users would like it	3.8%	10	Undesired
I can recover from mistakes quickly and easily	5.3%	8	Undesired
I can use it successfully every time	7.7%	5	Undesired

Table 7 Priority ranked criteria of Ease of Learning

Question	Priority	Rank	Desired or Undesired Data
I learned to use it quickly	25.2%	2	Desired
I easily remember how to use it	20.3%	3	Desired
It is easy to learn to use it	39.3%	1	Desired
I quickly became skilful with it	15.3%	4	Undesired

Table 8 Priority ranked criteria of satisfaction category

Question	Priority	Rank	Desired or Undesired Data
I am satisfied with it	19.9%	2	Desired
I would recommend it to a friend	5.4%	7	Undesired
It is fun to use	8.9%	5	Undesired
It works the way I want it to work	19.0%	3	Desired
It is wonderful	8.3%	6	Undesired
I feel I need to have it	12.5%	4	Undesired
It is pleasant to use	25.9%	1	Desired

comparisons for each expert to complete. Table 9 illustrates the data analysis conducted via the AHP online system used during this research and shows the rankings of the heuristics. Satisfaction was the most significant category with a priority of 13.7%, followed closely by usefulness, which had a priority of 13.6%. The least impactful heuristic as identified by the AHP analysis of this data was help and documentation (priority of 2.0%), closely followed by help users recognise, diagnose, and recover from errors (priority of 2.2%) and recognition rather than recall (priority of 2.3%).

4.4 Final version of proposed usability heuristics

The initiation process of the proposed usability heuristics is complete after identifying the rankings of the heuristics

involved via AHP. The rearrangement of these heuristics allows researchers to be aware of which aspects require more attention, thus improving the overall usability of the system. Table 10 displays the final version with the sub-heuristics present for each heuristic. However, the final step in this research is to validate it via heuristic evaluation of Microsoft Teams and Blackboard.

4.5 Validation of the proposed usability heuristics

One of the biggest flaws that accompany new heuristic proposals is the lack of validation and robustness (Hermawati and Lawson 2016). Without validation, the benefits of the proposed heuristics are unclear, and the overall reliability is affected. Therefore, this research completed a

Table 9 Rankings and priority of the proposed heuristics

Heuristics	Priority	Rank
Visibility of system status	11.0%	4
Match between system and the real world	3.5%	9
User control and freedom	3.4%	10
Consistency and standards	9.8%	6
Error prevention	2.6%	11
Recognition rather than recall	2.3%	12
Flexibility and efficiency of use	13.1%	3
Aesthetic and minimalist design	5.6%	8
Help users recognise, diagnose, and recover from errors	2.2%	13
Help and documentation	2.0%	14
Usefulness	13.6%	2
Ease of Use	11.0%	4
Ease of Learning	6.2%	7
Satisfaction	13.7%	1

Table 10 The final version with the sub-heuristics

Heuristics	Sub-Heuristics
Satisfaction	It is pleasant to use I am satisfied with it
Usefulness	It works the way I want it to work It is useful It helps me be more effective It helps me be more productive
Flexibility and efficiency of use	Provide accelerators like keyboard shortcuts and touch gestures Provide personalization by tailoring content and functionality for individual users Allow for customization, so users can make selections about how they want the product to work
Visibility of system status	Communicate clearly to users what the system's state is — no action with consequences to users should be taken without informing them Present feedback to the user as quickly as possible (ideally, immediately) Build trust through open and continuous communication
Ease of Use	It is flexible It is easy to use It is simple to use
Consistency and standards	Improve learnability by maintaining both types of consistency: internal and external Maintain consistency within a single product or a family of products (internal consistency) Follow established industry conventions (external consistency)
Ease of Learning	It is easy to learn to use it I learned to use it quickly I easily remember how to use it
Aesthetic and minimalist design	Keep the content and visual design of UI focus on the essentials Don't let unnecessary elements distract users from the information they really need Prioritize the content and features to support primary goals
Match between system and the real world	Ensure users can understand meaning without having to go look up a word's definition Never assume your understanding of words or concepts will match those of your users User research will help you uncover your users' familiar terminology, as well as their mental models around important concepts
User control and freedom	Support Undo and Redo Show a clear way to exit the current interaction, like a Cancel button Make sure the exit is clearly labelled and discoverable
Error prevention	Prioritize your effort: Prevent high-cost errors first, then little frustrations Avoid slips by providing helpful constraints and good defaults Prevent mistakes by removing memory burdens, supporting undo, and warning your users
Recognition rather than recall	Let people recognise information in the interface, rather than having to remember ("recall") it Offer help in context, instead of giving users a long tutorial to memorise Reduce the information that users have to remember
Help users recognise, diagnose, and recover from errors	Use traditional error message visuals, like bold, red text Tell users what went wrong in language they will understand — avoid technical jargon Offer users a solution, like a shortcut that can solve the error immediately
Help and documentation	Ensure that the help documentation is easy to search Whenever possible, present the documentation in context right at the moment that the user requires it List concrete steps to be carried out

validation process to clarify the advantages of the new usability heuristics and how it addresses the weak points of

previous heuristics. Heuristic evaluation allowed experts to identify 609 issues across both Microsoft Teams and

Blackboard via the new heuristics as displayed in Table 11. This is almost triple the amount of problems identified through Nielsen's heuristics. However, although Table 11 shows that the total number of problems identified (highlighted in bold) via the proposed heuristics is more than the total number of problems identified via Nielsen's heuristics (Table 4), we must pay attention to the first ten heuristics in the usability heuristics to make a direct and fair comparison. This comparison allows us to identify whether the sub-heuristics included have an effect on the evaluation of systems, since the first ten heuristics are the same as Nielsen's heuristics. The total number of problems recognised by the experts in the first ten heuristics amounted to 418, which is almost double the number of problems (238) identified via Nielsen's heuristics in the first part of this research. Therefore, this is a clear indication that the sub-heuristics provided more clarity for the experts in their evaluation and enabled them to identify more issues in the system than they usually would without the sub-heuristics.

4.6 Comparison against other heuristics

Since Nielsen's heuristics are applauded for simplicity, the researcher did not want to negate this integral characteristic and limited the sub-heuristics of the new proposal to three according to the level of importance so that it remains simple but effective during evaluation. Studies (Pierotti 1995) that extend Nielsen's heuristics drastically, disregard the importance of simplicity which is one of the reasons that made Nielsen's heuristics popular amongst evaluators. Furthermore, unlike other researchers (Paddison and

Englefield 2004) who use literature review and data analysis to develop and extend Nielsen's heuristics, this study believes that the most appropriate way to extend Nielsen's heuristics is with his own understanding of it as it is more reliable and is approved by Nielsen. Therefore, this research modified and extended Nielsen's heuristics by adding three sub-heuristics based on the updated tips of Nielsen's heuristics. In addition, many heuristics are specific to certain domains (Liao and Shen 2012; Dimaquila et al 2020), whereas our heuristics are for the general usability heuristic evaluation and can be broadly used. The researcher believes that the combination of Nielsen's heuristics and the USE Questionnaire provides the most essential heuristics that are integral for a successful system that ultimately helps developers build a robust and functional system.

5 Conclusion and future works

The aim of this study is to provide user interface designers with a new set of usability heuristics that provide more detail than Nielsen's heuristic evaluation, and to minimise the errors that can affect the use of the system. As a result, the researcher extended the Nielsen's usability heuristics by including four extra heuristics from the USE Questionnaire, where each heuristic contains three sub-heuristics gathered via AHP analysis. The results of AHP provide priority based rankings which helps evaluators identify which areas of a system require more attention so that efficiency and effectiveness are ensured during the

Table 11 Number of problems identified by using proposed Usability Heuristics

Usability Heuristics	Number of problems identified	
	Microsoft Teams	Blackboard
Visibility of system status	15	27
Match between system and the real world	18	25
User control and freedom	15	28
Consistency and standards	12	19
Error prevention	16	25
Recognition rather than recall	16	26
Flexibility and efficiency of use	21	30
Aesthetic and minimalist design	12	15
Help users recognise, diagnose, and recover from errors	17	34
Help and documentation	18	29
Usefulness	19	34
Ease of Use	19	31
Ease of Learning	22	29
Satisfaction	13	24
Total	233	376
	609	

development process to attain the main goal of user satisfaction. The final version of the usability heuristics proposal contained 14 heuristics, with each one containing three sub-heuristics, totalling 42 usability sub-heuristics. The methods undertaken to initiate these heuristics are in accordance with Hermawati and Lawson's (2016) conditions, since the researcher extracted information (tips and USE Questionnaire) and converted them into new heuristics and sub-heuristics in some cases. Two experiments (Nielsen's heuristic evaluation and the proposed usability heuristic evaluation), involving five experts, helped identify if the new usability heuristics detected more problems. Results show that the new usability heuristics helped experts identify almost triple the amount of issues compared to Nielsen's heuristics. AHP analysis shows that the highest priority ranked heuristic is satisfaction, closely followed by usefulness, and flexibility and efficiency of use. Evaluators should focus on these heuristics according to the usability heuristics in descending order, paying more attention to the heuristics that are higher up in the rankings. This research established and validated these heuristics via a literature review, modifying, extending, and merging Nielsen's heuristics and the USE Questionnaire, and finally, via basing them on empirical data (Ling and Salvendy 2005). Therefore, these usability heuristics have been thoroughly examined to ensure that it is legitimate and appropriate for evaluators to use.

One of the limitations of this study is that it tests the proposed usability heuristics with experts only, therefore future studies can present it to users and identify their opinion and other useful feedback to improve the findings. In addition, there are no female experts in this research and their insight might prove useful. Furthermore, the two tested systems were educational software's, thus future research can possibly attain more research information regarding other fields.

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